THE CLASS $\mathcal{G}$ BEYOND DEGREE FIVE: EXACT WITNESSES, THE 2-UNRAMIFIED CLASSIFICATION IN DEGREES 6–8, AND THE BOTTOM OF THE VARIANCE SPECTRUM

ABSTRACT. Let $\mathcal{G}$ be the class of totally real fields K with $2\mathcal{O}_K^{\succ 0} \subseteq \Sigma\mathcal{O}_K^2$. Its members of degree at most 5 are known (eight fields), and it is expected that there are no others. This note records a computational and theoretical attack on degrees 6, 7, 8 and on the structure of small-variance fields.

(i) We give an exact decision procedure for "$2\alpha \in \Sigma\mathcal{O}_K^2$" and a search for *exceptional* elements (witnesses), and run it on every totally real field of degree 3–8 in the LMFDB lists (193,467 cubic, 13,073 quartic, 9,494 quintic, 9,815 sextic, 301 septic and 920 octic fields). In degrees ≤ 5 it reproduces the Kala–Yatsyna classification exactly; in degrees 6, 7 and 8 every field has an explicit, independently certified witness.

(ii) We show that for fields in which 2 is unramified, membership in $\mathcal{G}$ implies every necessary condition (budget–cost, sharpened rigidity, square dichotomy, and their relative versions) used in the classification of totally real fields of degree 6–8 with a universal $\mathbb{Z}$-form. Combining this with that classification and with explicit $\mathcal{G}$-witnesses for all the fields that it has to construct (18 sextic and 394 octic fields; it constructs no septic field), we obtain: *no totally real field of degree 6, 7 or 8 in which 2 is unramified lies in $\mathcal{G}$.*

(iii) We determine the bottom of the variance spectrum. All totally real algebraic integers with $\mathrm{Var}(x) \leq 2$ are listed in degrees ≤ 10, and an auxiliary-function (linear programming) certificate, verified in interval arithmetic, shows that every other totally real algebraic integer has $\mathrm{Var}(x) \geq 1.705$. Consequently the values of Var below 1.705 are exactly $0, \frac{5}{4}, \frac{14}{9}, \frac{27}{16}$, and a totally real field has $\nu(K) < 1.705$ if and only if it contains $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\zeta_7)^+$. As a by-product, $\liminf_{\deg x\to\infty} \mathrm{Var}(x) \geq 1.705$, which improves the bound $\sqrt{e} = 1.6487\ldots$ quoted in the working notes.

(iv) For sextic fields with 2 ramified we show that the minimal-vector enumeration can be completed when the minimal vector generates K (about $4 \cdot 10^8$ leaves and 54,000 candidate minimal polynomials, all excluded by exact tests). Hence a sextic member of $\mathcal{G}$ must have 2 ramified and inherit its variance minimum from a subfield F which is one of $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\zeta_7)^+$, $\mathbb{Q}(\zeta_9)^+$, $\mathbb{Q}(\rho)$ (five of which are members of $\mathcal{G}$).

(v) We record shape and arithmetic-statistics data from the census, and explicit witnesses for the seven smallest non-member real cyclotomic fields left open by the dyadic criterion ($m = 36$, 44, 52, 76, 100, 108, 116).

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Date: November 5, 2026.

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1. Introduction

For a totally real field K of degree d write $\mathcal{O}_K^{\succ 0}$ for its totally positive integers and $\Sigma\mathcal{O}_K^2$ for the sums of squares. Put

$$\mathcal{G} = \{K \text{ totally real}: \ 2\mathcal{O}_K^{\succ 0} \subseteq \Sigma\mathcal{O}_K^2\}.$$

Kala and Yatsyna [5] proved that the members of degree at most 5 are

$$\mathbb{Q},\ \mathbb{Q}(\sqrt{2}),\ \mathbb{Q}(\sqrt{3}),\ \mathbb{Q}(\sqrt{5}),\ \mathbb{Q}(\sqrt{2},\sqrt{5}),\ \mathbb{Q}(\zeta_{20})^+,\ \mathbb{Q}(\zeta_7)^+,\ \mathbb{Q}(\rho) \quad (\rho^3 = 4\rho + 2), \tag{1}$$

and it is expected that (1) is all of $\mathcal{G}$. The working notes [10, 11, 12] develop the *trace variance* $\mathrm{Var}(x) = \tau(x^2) - \tau(x)^2$, $\tau = \frac{1}{d}\mathrm{Tr}_{K/\mathbb{Q}}$, as the natural height for this problem, and the companion repository [14] pushes the related *lifting problem* (universal $\mathbb{Z}$-forms) through degrees 6, 7 and 8 using the same invariants. The purpose of this note is to transfer as much as possible of that machinery to $\mathcal{G}$, to test it against complete field tables, and to understand the fields of small variance, which are the only places where members of $\mathcal{G}$ can live.

Throughout, $\Lambda_K = \mathcal{O}_K/\mathbb{Z}$ with the positive definite form Var is the *variance lattice* (up to scaling it is the shape of K), and

$$\nu(K) = \min_{x \in \mathcal{O}_K \setminus \mathbb{Z}} \mathrm{Var}(x), \qquad \nu_2(K) = \min_{x \in (\mathbb{Z}+\mathfrak{r}) \setminus \mathbb{Z}} \mathrm{Var}(x), \qquad t(K) = \min_{\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}} \tau(\alpha),$$

where $\mathfrak{r} = \prod_{\mathfrak{p}|2} \mathfrak{p}^{\lceil e_\mathfrak{p}/2 \rceil}$ is the dyadic square-root radical; one has $2\mathcal{O}_K \subseteq \mathfrak{r}$, $\nu \le \nu_2 \le 4\nu$, and $\nu_2 = 4\nu$ when 2 is unramified.

Results.

(1) *Anatomy of a representation* (§2). A complete description of what a representation $2\alpha = \sum x_i^2$ looks like in the low-trace range, valid without any hypothesis on 2. It contains the budget–cost inequality and rigidity of [10] and adds a *half-square dichotomy*: if $\tau(\alpha) < \nu(K)$ and $K \in \mathcal{G}$ then $2\alpha - c$ is a square in $\mathcal{O}_K$ for an integer $0 \le c < 2\min_j \sigma_j(\alpha)$.
(2) *An exact witness search* (§3) and a census of all LMFDB fields of degree 3–8 (§4); Theorem 4.1.
(3) *The 2-unramified classification in degrees* 6–8 (§5); Theorem 5.2.
(4) *The bottom of the variance spectrum* (§6); Theorems 6.1 and 6.2.
(5) *Sextic fields with* 2 *ramified* (§9): Theorem 9.6, the reduction to six inherited families.
(6) *Shapes and statistics* (§7), *residual cyclotomic fields* (§8), and corrections to the working notes (§10).

All computations use PARI/GP 2.15.4 [9]; the code and all outputs are in the repository accompanying this note (§11). Every exclusion of a field rests on an explicit element whose exceptional status is certified by exact arithmetic, independently of the (GRH-conditional) class group and unit computations used to *find* it.

2. Anatomy of a representation

Lemma 2.1. *Let K be totally real, $\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}$, and $2\alpha = \sum_{i=1}^{N} x_i^2$ with $x_i \in \mathcal{O}_K \setminus \{0\}$. Let $S = \{i : x_i \notin \mathbb{Z}\}$. Then $S \neq \emptyset$, $\sum_i x_i \in \mathfrak{r}$, and*

$$2\tau(\alpha) = \sum_{i \in S} \big(\mathrm{Var}(x_i) + \tau(x_i)^2\big) + \sum_{i \notin S} x_i^2. \tag{2}$$

Moreover:

1. *If $|S| = 1$, say $S = \{1\}$, put $c = \sum_{i \geq 2} x_i^2 \in \mathbb{Z}_{\geq 0}$. Then $2\alpha - c = x_1^2$, $c < 2\min_j \sigma_j(\alpha)$, $x_1 \in (c + \mathfrak{r}) \setminus \mathbb{Z}$, and $2\tau(\alpha) \geq \nu_2(K) + c$.*
2. *If $|S| = 2$, say $S = \{1, 2\}$, and $\tau(\alpha) < \frac{1}{2}\nu_2(K)$, then $x_2 = \varepsilon x_1 + k$ with $\varepsilon = \pm 1$, $k \in \mathbb{Z}$, and*

$$4\alpha = (2x_1 + \varepsilon k)^2 + \kappa, \qquad \kappa = k^2 + 2\sum_{i \geq 3} x_i^2 \in \mathbb{Z}_{\geq 0}, \quad \kappa \equiv 0, 3 \pmod 4,$$

   with $\tau(\alpha) \geq \mathrm{Var}(x_1) + \kappa/4$. If $\kappa = 0$ then $\alpha = x_1^2$.
3. *If $|S| \geq 3$ then $\tau(\alpha) \geq \frac{3}{2}\nu(K)$.*

Proof. If $S = \emptyset$ then $2\alpha \in \mathbb{Z}$, so $\alpha \in \mathbb{Z}$. The coupling $\sum_i x_i \in \mathfrak{r}$ is [10, dyadic coupling lemma]: $s = \sum x_i$ has $s^2 \equiv \sum x_i^2 \equiv 0 \pmod{2\mathcal{O}_K}$, so $v_{\mathfrak{p}}(s) \geq \lceil e_{\mathfrak{p}}/2 \rceil$ for every $\mathfrak{p} \mid 2$. Identity (2) is τ of the representation, with $\tau(x^2) = \mathrm{Var}(x) + \tau(x)^2$.

(1) $x_1^2 = 2\alpha - c$ is totally positive, so $c < 2\sigma_j(\alpha)$ for all j. For integers $n \equiv n^2 \pmod 2$, hence $\sum_{i \geq 2} x_i \equiv c \pmod 2$; as $2 \in \mathfrak{r}$, the coupling gives $x_1 \equiv -\sum_{i \geq 2} x_i \equiv c \pmod{\mathfrak{r}}$. Thus $x_1 \in (\mathbb{Z} + \mathfrak{r}) \setminus \mathbb{Z}$, so $\mathrm{Var}(x_1) \geq \nu_2$, and (2) gives $2\tau(\alpha) \geq \mathrm{Var}(x_1) + c$.

(2) Put $c_0 = \sum_{i \geq 3} x_i \in \mathbb{Z}$. The coupling gives $x_1 + x_2 \in -c_0 + \mathfrak{r}$, and $2x_2 \in \mathfrak{r}$ gives $x_1 - x_2 \in -c_0 + \mathfrak{r}$. If neither $x_1 + x_2$ nor $x_1 - x_2$ is rational, both have variance $\geq \nu_2$, and the parallelogram law for the quadratic form Var gives $2\nu_2 \leq \mathrm{Var}(x_1+x_2)+\mathrm{Var}(x_1-x_2) = 2\,\mathrm{Var}(x_1)+2\,\mathrm{Var}(x_2) \leq 2\tau(x_1^2)+2\tau(x_2^2) \leq 4\tau(\alpha)$, contrary to the hypothesis. Hence $x_2 = \varepsilon x_1 + k$. Expanding, $2\alpha = 2x_1^2 + 2\varepsilon k x_1 + k^2 + M$ with $M = \sum_{i \geq 3} x_i^2$, so $4\alpha = (2x_1 + \varepsilon k)^2 + k^2 + 2M$. Since $k^2 + M = 2(\alpha - x_1^2 - \varepsilon k x_1) \in 2\mathcal{O}_K \cap \mathbb{Z}$, $M \equiv k^2 \pmod 2$ and $\kappa \equiv 3k^2 \pmod 4$. Finally $\tau(\alpha) = \tau\big((x_1 + \varepsilon k/2)^2\big) + \kappa/4 \geq \mathrm{Var}(x_1) + \kappa/4$, and $\kappa = 0$ forces $k = 0 = M$, i.e. $\alpha = x_1^2$.

(3) Three summands of (2) are each $\geq \nu$. □

Corollary 2.2 ($\mathcal{G}$-dichotomies)**.** *Let $K \in \mathcal{G}$, $K \neq \mathbb{Q}$, and $\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}$. Write $\nu = \nu(K)$, $\nu_2 = \nu_2(K)$.*

1. (budget–cost [10, budget–cost inequality]) $\tau(\alpha) \geq \min(\nu, \frac{1}{2}\nu_2)$.
2. (half-square dichotomy) *If $\tau(\alpha) < \nu$, then $2\alpha - c = y^2$ for some $c \in \mathbb{Z}$ with $0 \leq c < 2\min_j \sigma_j(\alpha)$ and some $y \in (c + \mathfrak{r}) \setminus \mathbb{Z}$; moreover $2\tau(\alpha) \geq \nu_2 + c$.*
3. (square window) *If $\tau(\alpha) < \min(\frac{3}{2}\nu,\ \nu + \frac{3}{4},\ \frac{1}{2}\nu_2)$, then α is a square in $\mathcal{O}_K$.*
4. (witness by trace) *If $\tau(\alpha) < \min(\nu, \frac{1}{2}\nu_2)$ for some $\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}$ of an arbitrary totally real K, then $2\alpha \notin \Sigma\mathcal{O}_K^2$; in particular $K \notin \mathcal{G}$, and α itself is a witness.*

Proof. (1), (2), (4) are immediate from Lemma 2.1. For (3): $|S| \geq 3$ is excluded by $\tau < \frac{3}{2}\nu$, and $|S| = 1$ by $\tau < \frac{1}{2}\nu_2$ (Lemma 2.1(1)); so $|S| = 2$, and $\nu + \kappa/4 \leq \tau(\alpha) < \nu + \frac{3}{4}$ forces $\kappa \in \{0, 1, 2\}$, hence $\kappa = 0$ by the congruence, hence $\alpha = x_1^2$. □

Remark 2.3. Apply (2) to the *lower rung* $\alpha_- = x-n$ of any $x \in \mathcal{O}_K \setminus \mathbb{Z}$, where $n = \lceil \theta_1 \rceil - 1$ and θ_1 is the least conjugate of x. Then $\min_j \sigma_j(\alpha_-) = \theta_1 - n \in (0,1]$, so $c \in \{0,1\}$: *if* $\tau(x-n) < \nu$ *then* $2x - 2n - c$ *is a square in* $\mathcal{O}_K$ *for some* $c \in \{0,1\}$. This is the $\mathcal{G}$-analogue of the "rung split" of [14, degree-8 rung split], and it is the only way in which the lower rung can be cheap. The same holds for the upper rung $\lfloor \theta_d \rfloor + 1 - x$.

2.1. **The unramified case, and relative versions.** If 2 is unramified then $\mathfrak{r} = 2\mathcal{O}_K$, $(\mathbb{Z} + \mathfrak{r})/\mathbb{Z} = 2\Lambda_K$, and $\nu_2 = 4\nu$. Then Corollary 2.2(1) reads $\tau(\alpha) \geq \nu$, the half-square alternative never occurs (it forces $\tau(\alpha) \geq 2\nu$), and the window of (3) is $\min(\frac{3}{2}\nu, \nu + \frac{3}{4})$.

Let $F \subsetneq K$ be a subfield, $\mathrm{Var}_{K/F}$ the relative variance (an F-valued totally nonnegative quadratic form vanishing exactly on F), and $\nu_{K/F} = \min\{\tau_F(\mathrm{Var}_{K/F}\, x) : x \in \mathcal{O}_K \setminus \mathcal{O}_F\}$; by the law of total variance [10, ANOVA identity], $\tau(x^2) \geq \tau_F(\mathrm{Var}_{K/F}\, x)$.

Proposition 2.4 (relative version, 2 unramified)**.** *Let* $K \in \mathcal{G}$ *with* 2 *unramified in* K, $F \subsetneq K$, *and* $\alpha \in \mathcal{O}_K^{\succ 0} \setminus F$. *Then* $\tau(\alpha) \geq \nu_{K/F}$. *If moreover* $\tau(\alpha) < \frac{3}{2}\nu_{K/F}$, *then* $\alpha = y^2 + \gamma y + \beta$ *with* $y \in \mathcal{O}_K \setminus \mathcal{O}_F$, $\beta, \gamma \in \mathcal{O}_F$, $\tau_F(\mathrm{Var}_{K/F}\, y) \leq \tau(\alpha)$, *and* $4\beta - \gamma^2$ *totally nonnegative.*

Proof. Let $S_F = \{i : x_i \notin F\} \neq \emptyset$. If $|S_F| \geq 2$ then $\tau(\alpha) \geq \nu_{K/F}$; if $|S_F| \geq 3$ then $\tau(\alpha) \geq \frac{3}{2}\nu_{K/F}$. If $S_F = \{1\}$, the coupling (with $\mathfrak{r} = 2\mathcal{O}_K$) gives $x_1 \in \mathcal{O}_F + 2\mathcal{O}_K$, so $x_1 = f + 2w$ with $w \notin F$ and $\tau(x_1^2) \geq 4\nu_{K/F}$, i.e. $\tau(\alpha) \geq 2\nu_{K/F}$. If $S_F = \{1,2\}$, then $x_1 \pm x_2 \in \mathcal{O}_F + 2\mathcal{O}_K$, and if both are outside F the parallelogram law for $\tau_F \circ \mathrm{Var}_{K/F}$ gives $\tau(\alpha) \geq 2\nu_{K/F}$. Otherwise $x_2 = \pm x_1 + \gamma$ with $\gamma \in \mathcal{O}_F$; then $\alpha = x_1^2 \pm \gamma x_1 + \beta$ with $2\beta = \gamma^2 + \sum_{i \geq 3} x_i^2$, $\beta \in \mathcal{O}_K \cap F = \mathcal{O}_F$, and $4\beta - \gamma^2 = \gamma^2 + 2\sum_{i \geq 3} x_i^2 \succeq 0$. □

3. An exact decision procedure, and certificates

3.1. **Deciding** $\beta \in \Sigma\mathcal{O}_K^2$**.** Let $\beta \in \mathcal{O}_K^{\succ 0}$. Every summand x of a representation $\beta = \sum x_i^2$ satisfies $\sigma_j(x)^2 \leq \sigma_j(\beta)$ for all j, hence lies in the finite set

$$X(\beta) = \Big\{ x \in \mathcal{O}_K \setminus \{0\} : \ \textstyle\sum_j \sigma_j(x)^2/\sigma_j(\beta) \leq d \Big\} \cap \{x : \beta - x^2 \succeq 0\}.$$

We enumerate the ellipsoid by Fincke–Pohst (PARI's `qfminim` on the weighted Gram matrix, LLL-reduced, at a working precision adapted to the spread of the conjugates of β) and test $\beta - x^2 \succeq 0$ exactly. A depth-first search over multisets, with a memo on remainders and a direct look-up for one and two squares, decides whether $\beta \in \Sigma\mathcal{O}_K^2$. Every representation found is re-verified by exact multiplication.

3.2. **Rigorous signs.** All decisions about signs of conjugates are exact. The embedding matrix is computed to ≥ 100 digits, and a numerical sign is accepted only when the computed value exceeds an explicit bound $(\|x\|_1 + 1)\max|M_{ij}|\,10^{-90}$ for the error; otherwise the sign is decided from a certified high-precision evaluation (precision doubled until $|v|$ exceeds a rigorous evaluation error) or by a Sturm sequence on the characteristic polynomial. This matters: in large-regulator fields the totally positive generators of small ideals are unavoidably unbalanced (conjugates of size $10^{\pm 50}$ occur for cubic fields of discriminant $\approx 2 \cdot 10^6$), and a first version of the code that used a fixed tolerance produced wrong sign vectors of fundamental units there. The bug was caught by the final certification step below.

3.3. **The witness search.** For a field K we try, in order: (U) the *unit criterion*: every $u \in U_K^+$ with $u, 2u \notin K^{\times 2}$ is a witness, since $\mathrm{N}(u) = 1$ allows at most two squares and equality in O'Meara's superadditivity forces u or $2u$ to be a square [10, unit dichotomy lemma]; (T) all $\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}$ with $\tau(\alpha) \leq T$, enumerated in Λ_K using $\mathrm{Var}(\alpha) \leq (d-1)\tau(\alpha)^2$ and the least totally positive shift of each class (Corollary 2.2(4) explains why such elements are natural witnesses);

TABLE 1. The census. U, T, N: number of fields whose first witness is a unit, a small-trace element, or a small-norm generator. "max N" is the largest norm of a first witness of type N.

d	fields	max d_K	U	T	N	left	$h^+ = h$	max N
3	193467	$3.37 \cdot 10^6$	112091	1	81373	2 (49,148)	42.1%	197
4	13073	$1.00 \cdot 10^6$	7113	1118	4840	2 (1600,2000)	30.7%	81
5	9494	$1.00 \cdot 10^7$	5170	3097	1227	0	45.5%	17
6	9815	$1.00 \cdot 10^8$	5024	4413	378	0	40.6%	43
7	301	$2.14 \cdot 10^8$	156	145	0	0	48.2%	–
8	920	$6.97 \cdot 10^9$	405	515	0	0	51.1%	–

(N) for every ideal of norm $\leq N_{\max}$ that is narrowly principal, every totally positive generator modulo U_K^2, reduced in trace by squares of units.

3.4. **Certificates.** A witness α is accepted only after (a) total positivity is re-proved by a Sturm sequence of its characteristic polynomial, with no floating point, and (b) the decision procedure is rerun from scratch on 2α and finds no representation. Neither step uses the class group or the units, so every exclusion below is unconditional, although the search that found the witness used `bnfinit` (correct under GRH).

4. THE CENSUS IN DEGREES 3–8

We ran the search on the LMFDB lists of totally real fields [6] as mirrored in [15] (the lists used by the `sails` project). The bounds were $T = 3$ ($T = 2$ in degree 3) and $N_{\max} = 64, 128, 300, 300, 400$ in degrees $4, 5, 6, 7, 8$ ($N_{\max} = 64$ in degree 3); the few survivors were rerun with $N_{\max} = 1500$. One cubic field ($d_K = 3{,}307{,}164$, regulator ≈ 1288) needed a separate run: its prime of norm 2 only has totally positive generators of size 10^{54}, and we certified instead a witness of norm 3.

Theorem 4.1. *Every one of the 9815 fields of degree 6 ($d_K \leq 1.00 \cdot 10^8$), 301 fields of degree 7 ($d_K \leq 2.14 \cdot 10^8$), 920 fields of degree 8 ($d_K \leq 6.97 \cdot 10^9$) in the LMFDB lists has an explicit element $\alpha \in \mathcal{O}_K^{\succ 0}$ with $2\alpha \notin \Sigma\mathcal{O}_K^2$, certified as in §3; none of these fields lies in $\mathcal{G}$. In degrees 3, 4, 5 the fields without a witness are exactly the members of (1) in the lists.*

Remark 4.2 (validation against [5])**.** In degrees 3, 4, 5 the procedure finds a witness for every field except exactly the members in (1) (discriminants $49, 148$; $1600, 2000$; none), which survive up to $N_{\max} = 1500$. Since our implementation shares no code with [5], this is an independent check of their classification on these lists, and conversely their theorem is a check of our code: a single false witness or a missed representation in a member would have shown up here.

Remark 4.3 (the extreme cases)**.** The fields that need the largest witnesses are imprimitive fields with a subfield of very small variance and a large discriminant: the seven quartic fields with $d_K \in [5.7 \cdot 10^5, 10^6]$ that survive $N_{\max} = 64$ all contain $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\sqrt{2})$, have normalised shape minimum ≈ 0.09 (against a median 0.58), and need witnesses of norm 71–81. This is the "inherited witness" phenomenon of [10, "What remains, and why"] seen in data: a small-variance subfield blocks every trace-based witness, and only norm-based witnesses remain, whose size grows with d_K.

TABLE 2. $\mathcal{G}$-witnesses for the fields left by the pipeline of [14] that have 2 unramified, in degree 6, and for the two octic fields excluded there only by (N).

d_K	witness α (minimal polynomial)	$\mathrm{N}_{K/\mathbb{Q}}\,\alpha$
300125	$x^6-14x^5+73x^4-182x^3+227x^2-133x+29$	29
485125	$x^6-13x^5+64x^4-151x^3+177x^2-96x+19$	19
820125	x^3-6x^2+9x-3	9
1416125	$x^6-14x^5+75x^4-191x^3+232x^2-117x+19$	19
3162625	$x^6-15x^5+86x^4-234x^3+299x^2-142x+4$	4
442050625	$x^4-9x^3+27x^2-31x+11$	121
656505625	$x^4-9x^3+27x^2-31x+11$	121

5. FIELDS WITH 2 UNRAMIFIED IN DEGREES 6, 7, 8

The repository [14] proves that no totally real field of degree 6, 7 or 8 admits a universal $\mathbb{Z}$-form. Its proof uses the following necessary conditions on a field K (we refer to the notes in [14] for the precise statements):

(BC) $\tau(\alpha)\geq\nu(K)$ for all $\alpha\in\mathcal{O}_K^{\succ 0}\setminus\mathbb{Z}$, and its integer-part consequences (downspread, the quadratic filter);

(RG) rigidity: $\tau(\alpha)<\frac{3}{2}\nu(K)$ implies $\alpha=y^2+cy+b$ with $y\notin\mathbb{Z}$, $\mathrm{Var}(y)\leq\tau(\alpha)$, $\kappa=4b-c^2\geq 0$, $\kappa\equiv 0,3 \pmod 4$; in particular the square dichotomy $\tau(\alpha)<\min(\frac{3}{2}\nu,\nu+\frac{3}{4})\Rightarrow\alpha\in\mathcal{O}_K^2$;

(REL) the relative versions of (BC) and (RG) over every subfield;

and, only at the very end, condition (N) at explicit elements.

Proposition 5.1. *If $K\in\mathcal{G}$ and 2 is unramified in K, then K satisfies (BC), (RG) and (REL).*

Proof. (BC) and (RG) are Corollary 2.2 with $\nu_2=4\nu$ together with Lemma 2.1(2) (which gives exactly the shape y^2+cy+b with $\kappa=k^2+2\sum_{i\geq 3}x_i^2\geq 0$); (REL) is Proposition 2.4. □

Theorem 5.2. *No totally real field of degree 6, 7 or 8 in which 2 is unramified belongs to $\mathcal{G}$.*

Proof. By Proposition 5.1, such a field passes every test of [14] that precedes the final condition-(N) step. That computation leaves, in degree 6, the 18 fields of [14, degree-6 Case B] (Case B; Case A is empty); in degree 7, nothing; in degree 8, the four fields of Case A and the 390 fields of Case B over $\mathbb{Q}(\sqrt{5})$. For each of these $18+4+390$ fields we found an explicit $\alpha\in\mathcal{O}_K^{\succ 0}$ with $2\alpha\notin\Sigma\mathcal{O}_K^2$, certified as in §3 (files `results/g2o_d6.txt`, `g2o_d8A.txt`, `g2o_d8B.txt`); Table 2 lists the ones with 2 unramified in degree 6 and the two octic fields that survive all tests of [14] except (N). □

Remark 5.3. (a) The proof uses the computations of [14] as a black box; we have checked that each exclusion test there is one of (BC), (RG), (REL). We reran its degree-6 and degree-7 pipelines from scratch: the outputs (enumeration counts, the status of every candidate, the relative enumerations and the final list of 18 sextic fields) are identical to the published ones. The degree-8 computation (about three hours) was not rerun. (b) Most of the fields in the list have 2 ramified; they are not covered by Proposition 5.1, but our witnesses exclude them from $\mathcal{G}$ anyway. (c) For $\mathbb{Q}(\sqrt{5},\zeta_7^+)$ ($d_K=300125$) and the two octic fields, the $\mathcal{G}$-witness is the same element at which (N) fails in [14]: the element of trace 14 and norm 29, and the element of trace 18 in K (trace 9 and norm 11 in the quartic subfield of discriminant 725), respectively.

TABLE 3. The variance spectrum below 1.9 in degrees ≤ 10: value, degrees in which it occurs, number of polynomials (up to $x \mapsto \pm x + k$).

Var	decimal	degrees	#
5/4	1.2500	2	1
14/9	1.5556	3	1
27/16	1.6875	4	1
7/4	1.7500	4,8	3
44/25	1.7600	5	1
16/9	1.7778	9	1
9/5	1.8000	10	3
65/36	1.8056	6	2
90/49	1.8367	7	3
46/25	1.8400	5,10	9
37/20	1.8500	10	12
119/64	1.8594	8	8
152/81	1.8765	9	15
17/9	1.8889	6	4
189/100	1.8900	10	43

6. The bottom of the variance spectrum

6.1. **Enumeration.** Up to $x \mapsto \pm x + k$ and conjugation, a totally real algebraic integer of degree e is a real-rooted monic $f \in \mathbb{Z}[t]$ with $a_1 \in \{0, -1, \dots, -\lfloor e/2 \rfloor\}$, and $e\,\mathrm{Var}(x) = \frac{e-1}{e} a_1^2 - 2a_2$. We enumerated all such f with $\mathrm{Var} \le 2$ (Robinson's method, the enumerator of [14] in its unfiltered mode) for $e \le 10$ and kept the irreducible ones.

6.2. **Auxiliary functions.** Let $I = [s_0, s_1] \subseteq [0, \frac{1}{2}]$, $Q_1, \dots, Q_k \in \mathbb{Z}[t]$ and $c_1, \dots, c_k \ge 0$ be such that

$$f_I(t) = \mathrm{dist}(t, I)^2 - \sum_i c_i \log |Q_i(t)| \;\ge\; m_I \qquad (t \in \mathbb{R}). \tag{3}$$

If x is a totally real algebraic integer with $\tau(x) \in I$ and $Q_i(x) \ne 0$ for all i, then averaging (3) over the conjugates of x and using $|\,\mathrm{N}(Q_i(x))| \ge 1$ gives $\mathrm{Var}(x) \ge \frac{1}{e}\sum_j \mathrm{dist}(\theta_j, I)^2 \ge m_I$. Every x can be normalised by $x \mapsto \pm x + k$ to have $\tau(x) \in [0, \frac{1}{2}]$.

We took for the Q_i the linear polynomials $t - k$ and all integer shifts and reflections, with roots in $[-4.5, 5]$, of the irreducible polynomials of degree ≤ 8 found in the enumeration (3,068 polynomials; 959 for degree ≤ 7), solved the linear program for the c_i on a grid, rounded the c_i down to rationals, lowered each m_I by 10^{-3}, and *verified* (3) on all of $\mathbb{R}$, for each of the 10 intervals $I = [\frac{k}{20}, \frac{k+1}{20}]$ covering $[0, \frac{1}{2}]$: on $[-7, 7.5]$ by interval arithmetic (`mpmath.iv`, 120 bits) with adaptive subdivision, using $\log|Q_i(t)| \le \log \max_{\mathrm{cell}} |Q_i|$; outside, by the explicit increasing minorant $(|t| - \frac{1}{2})^2 - \sum_i c_i(\log \|Q_i\|_1 + \deg Q_i \log|t|)$.

Theorem 6.1. *Let $m^* = \frac{170577}{100000} = 1.70577$. There is an explicit set $\mathcal{Q}$ of 122 irreducible polynomials in $\mathbb{Z}[t]$, of degrees ≤ 8 (file `results/lp/cert_deg8.json`), such that every totally real algebraic integer x which is not, after $x \mapsto \pm x + k$ ($k \in \mathbb{Z}$), a root of a member of $\mathcal{Q}$ satisfies*

$$\mathrm{Var}(x) \;\ge\; m^*.$$

The members of $\mathcal{Q}$ have degree ≤ 8, so their roots appear in the enumeration, and among them only the following have variance $< m^*$: $\frac{1+\sqrt{5}}{2}$ ($\mathrm{Var} = \frac{5}{4}$), $2\cos\frac{2\pi}{7}$ ($\frac{14}{9}$), and a root of $t^4 - t^3 -$

$3t^2+t+1$ ($\frac{27}{16}$; *it generates the totally real quartic field of discriminant* 725, *the one of least discriminant). Consequently*

$$\{\mathrm{Var}(x) : x \text{ a totally real algebraic integer}\} \cap [0, m^*) = \{0, \tfrac{5}{4}, \tfrac{14}{9}, \tfrac{27}{16}\},$$

each value being attained by a unique integer up to $x \mapsto \pm x+k$ *and conjugation, and* $\mathrm{Var}(x) \geq m^*$ *for every totally real algebraic integer of degree* ≥ 5. *With the* 109 *polynomials of degree* ≤ 7 *of* `cert_deg7.json` *one gets* $\frac{33839}{20000} = 1.69195$, *and with the* 36 *of degree* ≤ 5 *of* `cert_deg5.json` *already* $\frac{32597}{20000} = 1.62985$.

Theorem 6.2. *Let* K *be totally real and* m^* *as in Theorem 6.1. Then* $\nu(K) < m^*$ *if and only if* K *contains* $\mathbb{Q}(\sqrt{5})$ *or* $\mathbb{Q}(\zeta_7)^+$; *and then* $\nu(K) = \frac{5}{4}$ *if* $\sqrt{5} \in K$, *and* $\nu(K) = \frac{14}{9}$ *otherwise. In particular* $\nu(K) \geq m^*$ *for every totally real field without a quadratic or cubic subfield (for instance every primitive field of degree* ≥ 4, *and every field of degree prime to* 6*).*

Proof. Let $x \in \mathcal{O}_K \setminus \mathbb{Z}$ realise $\nu(K)$. If $\nu(K) < m^*$, Theorem 6.1 identifies x, up to $x \mapsto \pm x + k$ and conjugation, as the golden ratio, $2\cos\frac{2\pi}{7}$, or a root of $t^4 - t^3 - 3t^2 + t + 1$. The last generates the quartic field F_{725} of discriminant $725 = 5^2 \cdot 29$, which contains $\mathbb{Q}(\sqrt{5})$; so in that case already $\nu(K) \leq \frac{5}{4}$, and the value $\frac{27}{16}$ never occurs as $\nu(K)$. Conversely $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\zeta_7)^+$ contain elements of variance $\frac{5}{4}$ and $\frac{14}{9}$, and $\frac{5}{4}$ is the absolute minimum [10, sharp floor]. □

Remark 6.3. The exhaustive enumeration also lists the fields generated by integers of variance ≤ 2 (file `results/spectrum/fields_Vle2.txt`): there are $2, 2, 4, 4, 18, 43, 195, 463, 734$ of them in degrees $2, \ldots, 10$ (of which $2, 2, 1, 4, 7, 43, 150, 453, 650$ primitive). Any totally real field with $\nu(K) \leq 2$ whose minimal vector has degree $e \leq 10$ contains one of them. Their number grows quickly with the degree, in line with $\liminf \mathrm{Var} \leq \lambda_{\mathrm{SSS}} < 2$.

Remark 6.4. (a) Theorem 6.1 gives $\liminf_{\deg x \to \infty} \mathrm{Var}(x) \geq m^*$; the working notes [10, the limiting-variance theorem] quote $\sqrt{e} = 1.6487\ldots$ (Schur's argument via Hermite discriminants; see also [3, 2]), and the upper bound $\lambda_{\mathrm{SSS}} \leq 1.8984$ comes from Smith's realisation theorem [13, 8]. We have not found a better lower bound in the literature, but have not made an exhaustive search. (b) The degree-≤ 5 certificate already gives $m^* = 1.6298$, enough to show that $\frac{5}{4}$ and $\frac{14}{9}$ are the only nonzero values below 1.6298; this sharpens the "bottom of the spectrum" theorem of [10, bottom of the spectrum], which proves that $(\frac{5}{4}, \frac{14}{9})$ contains no value. (c) The quartic field of discriminant 725 is precisely the subfield on which the two octic survivors of [14] fail (N) and $\mathcal{G}$ (Table 2): small-variance fields are where the hard cases live.

6.3. **Consequences for $\mathcal{G}$.** The eight members (1) have $\nu \in \{\frac{5}{4}, \frac{14}{9}, 2, \frac{20}{9}, 3\}$. By Theorem 6.2, a field $K \in \mathcal{G}$ with $\nu(K) < m^*$ contains $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\zeta_7)^+$; every other member has $\nu(K) \geq m^*$, and by the degree-free exclusion [10, degree-free exclusion] its minimal vector has degree e with $m^* \leq \nu(K) \leq 2e + 2\sqrt{e^2 - 1}$.

7. Shapes and arithmetic statistics

The lattice $(\Lambda_K, \mathrm{Var})$ is the shape of K up to scaling [7], with determinant d_K/d^d. We record the *normalised shape minimum* $\hat{\nu}(K) = \nu(K)\,(d^d/d_K)^{1/(d-1)} \in (0, \gamma_{d-1}]$.

Table 4 summarises the shapes. Three features stand out. (a) In the prime degrees 5 and 7 the shape minima concentrate in a band around 0.7–1.0, well below the Hermite bound, while in degrees 4, 6, 8 there is a heavy lower tail: these are the imprimitive fields, whose minimal vector lies in a subfield of small variance and whose shape is therefore skewed. (b) In degree 6, witnesses of type T (small trace, i.e. $t(K)$ small compared with ν and ν_2: Corollary 2.2(4)) occur for well-rounded shapes and witnesses of type N for skewed ones: the medians of $\hat{\nu}$ are 0.77 and 0.37. This is the budget–cost mechanism seen statistically: a round shape forces ν to be large

TABLE 4. Normalised shape minimum $\hat{\nu}$ in the census: quantiles, the Hermite bound γ_{d-1}, medians by type of first witness, and the share of fields with $h^+ = h$ (equivalently $U_K^+ = U_K^2$).

d	$q_{0.01}$	$q_{0.1}$	$q_{0.5}$	$q_{0.9}$	max	γ_{d-1}	U	T	N	$h^+ = h$
3	0.19	0.27	0.62	0.96	1.15	1.15	0.61	1.15	0.63	42.1%
4	0.08	0.12	0.58	0.90	1.19	1.26	0.55	0.58	0.62	30.7%
5	0.47	0.55	0.73	0.94	1.20	1.41	0.73	0.72	0.80	45.5%
6	0.27	0.34	0.74	0.95	1.32	1.52	0.74	0.77	0.37	40.6%
7	0.85	0.88	0.97	1.08	1.25	1.67	0.96	0.98	–	48.2%
8	0.53	0.55	0.92	1.08	1.28	1.81	0.89	0.95	–	51.1%

TABLE 5. Cubic fields: empirical distribution of $\hat{\nu} = 1/\operatorname{Im} z$ (the shape z in the modular fundamental domain) against the hyperbolic-measure prediction $P(\hat{\nu} < s) = 3s/\pi$ ($s \le 1$) of Bhargava–Harron equidistribution.

s	$3s/\pi$	all	$d_K \le 10^5$	$d_K > 10^6$
0.2	0.191	0.023	0.000	0.032
0.3	0.286	0.130	0.000	0.142
0.4	0.382	0.249	0.111	0.264
0.5	0.477	0.367	0.230	0.380
0.6	0.573	0.479	0.370	0.487
0.7	0.668	0.598	0.505	0.607
0.8	0.764	0.715	0.652	0.722
0.9	0.859	0.829	0.794	0.831
1.0	0.955	0.944	0.926	0.945

compared with the cheapest totally positive element, while a skewed (imprimitive) shape inherits a small ν from a subfield and only a norm-type witness is left. In degree 4 the distinction is not visible. (c) Between one half and two thirds of the fields have $U_K^+ \neq U_K^2$; in odd degree all of these are excluded by the unit criterion, since then (2) is not the square of an ideal and $2u$ can never be a square. The proportion of fields with $h^+ = h$ that we observe (31–48% in degrees 4–7) is the kind of statistic predicted by the heuristics of Dummit and Voight for signature ranks of units [4]; we have not compared the numbers with their predictions.

For cubic fields the shape is a point z of the modular curve and $\hat{\nu} = 1/\operatorname{Im} z$, so Bhargava–Harron equidistribution predicts $P(\hat{\nu} < s) = 3s/\pi$ for $s \le 1$ (Table 5). The deep cusp is, however, arithmetically forbidden: a generator x of a cubic field has $d_K \le \operatorname{disc}(x) \le (3 \operatorname{Var} x)^3$, so $\nu(K) \ge d_K^{1/3}/3$ and $\hat{\nu} \ge \sqrt{3}\, d_K^{-1/6}$. This lower cut-off is what makes the set of cubic fields with bounded ν finite, although equidistribution alone would allow $\asymp \sqrt{X}$ of them below discriminant X; it is the shape-theoretic form of the effective Northcott theorem of [10]. In the data the cusp is visibly depleted well beyond the cut-off ($P(\hat{\nu} < \frac{1}{2})$ is 0.23 for $d_K \le 10^5$ and 0.38 for $d_K > 10^6$, against the limit 0.48), so the approach to equidistribution near the cusp is slow, as one expects from a secondary term; this is not in conflict with [1], which is asymptotic.

Table 6 isolates the *inherited* families, the sextic fields that contain a member of $\mathcal{G}$ (or have the same ν as one). They are the extreme cases of the census: they pass the unit and trace tests far more often than average, and the fields of largest discriminant in the $\mathbb{Q}(\zeta_7)^+$ family need the largest norm witnesses of all sextic fields ($\mathrm{N}\,\alpha = 43$ at $d_K \approx 9 \cdot 10^7$). This

TABLE 6. Sextic fields of the census whose ν is that of a member of $\mathcal{G}$ (by Theorem 6.2 the first two rows are exactly the fields containing $\mathbb{Q}(\sqrt{5})$, resp. $\mathbb{Q}(\zeta_7)^+$ but not $\sqrt{5}$): number of fields, types of first witness, and the norms of the type-N witnesses.

family	fields	U	T	N	median N	max N
$\sqrt{5} \in K$ $(\nu = \frac{5}{4})$	1270	710	446	114	4	19
$\mathbb{Q}(\zeta_7)^+ \subseteq K$ $(\nu = \frac{14}{9})$	491	269	53	169	8	43
$\nu = 2$	563	298	254	11	2	17
$\nu = \frac{20}{9}$	250	92	112	46	7	34
$\nu = 3$	299	130	168	1	6	6

is consistent with the reduction of §9, where the cases left open are exactly extensions of $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\zeta_7)^+, \mathbb{Q}(\zeta_9)^+, \mathbb{Q}(\rho)$.

8. The residual real cyclotomic fields

The dyadic criterion of [11] leaves, among $m = 4m_0$, the family with $-1 \in \langle 2 \rangle$ modulo m_0 and $\varphi(m_0)/\operatorname{ord}_{m_0}(2)$ odd; for $m_0 \le 99$ these are $m_0 = 3$, 5, 9, 11, 13, 19, 25, 27, 29, 37, 43, 53, 59, 61, 67, 81, 83 (we re-derived this table). The first two are members. Our search gives explicit witnesses for the next ones:

m	m_0	$[F_m : \mathbb{Q}]$	type	$\mathrm{N}\,\alpha$	$\mathrm{Tr}\,\alpha$
36	9	6	trace	24	18
44	11	10	norm	89	28
52	13	12	norm	13	52
76	19	18	norm	229	106
100	25	20	norm	5	120
108	27	18	norm	109	108
116	29	28	unit	1	120

These exclusions are unconditional: each witness is certified as in §3, with no class-group or GRH input. They are residual for the dyadic criterion of [11]; the working notes [10] state that the conductors $4q$ with $q \le 37$ are settled by the signature law and that all conductors not of the form $4q$ are settled as well, so for these m our witnesses are an independent, explicit confirmation rather than new exclusions. (F_{116} has $h^+/h = 16$, so it is also excluded by the unit criterion of [10], and indeed its witness is a unit.) For F_{52} the witness has norm 13: it is a totally positive generator of the ramified prime above 13, as predicted by the signature law ($13 \equiv 1 \pmod 4$ and $4 \mid \operatorname{ord}_{13} 2 = 12$), and the exact decision procedure replaces the analytic non-representation criterion, which needs $q \ge 269$.

9. The sextic problem with 2 ramified

By Theorem 5.2 and the census, a sextic member of $\mathcal{G}$ would have 2 ramified and discriminant beyond the LMFDB list. We record how the minimal-vector method splits in that case. Let x_1 realise ν, with rungs $\alpha_\pm$ as in Remark 2.3.

Regime 1: $\nu_2 \ge 2\nu$. Then Corollary 2.2(1) gives $\tau(\alpha) \ge \nu$ for every $\alpha \in \mathcal{O}_K^{\succ 0} \setminus \mathbb{Z}$, so the downspread bound $d\nu \ge 2(\nu-1)^2$ and the quadratic filter hold, and Case A of the enumeration of [14] (with its 15,720 irreducible candidates, $\nu \le 4.79$) covers all such K with $\mathbb{Q}(x_1) = K$.

What is lost is rigidity, whose window shrinks to $\tau < \frac{1}{2}\nu_2$ when $\nu_2 < 3\nu$; the 9,785 candidates that [14] excludes by rigidity must instead be tested at the level of the field.

Regime 2: $\nu \le \nu_2 < 2\nu$. Budget–cost only gives $\tau \ge \frac{1}{2}\nu_2$. A cheap rung is then a *half-square*: $2x_1 - 2n - c = y^2$ with $c \in \{0,1\}$ (Remark 2.3). If one rung is regular, the downspread argument gives $\frac{3}{2}\nu \le \sqrt{2d\nu}+2$, i.e. $\nu \le 7.77$; if both are half-squares, $y^2+z^2 = N$ for an integer $N \in [2\nu, 4\nu)$ and $\nu \le 15.75$. In both cases one enumerates y with $p_2(y) = d(2\tau(\alpha_-) - c)$ and $p_4(y) = d\big(4\nu + (2\tau(\alpha_-) - c)^2\big)$ fixed, as in the rung split of [14, degree-8 rung split].

Case B (the minimal vector lies in a proper subfield F). If $[K : F]$ is odd, an element of F that is a square in K is already a square in F (otherwise $F(\sqrt{\beta}) \subseteq K$ would have even degree over F). Hence every half-square condition of Corollary 2.2(2) for elements of F must already hold in F, with $\nu = \nu(K) = \nu(F)$.

Proposition 9.1. *Let $K \in \mathcal{G}$, and suppose that a minimal vector of Λ_K lies in a quadratic subfield F with $[K : F]$ odd. Then $F \in \{\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5})\}$.*

Proof. Write $F = \mathbb{Q}(\sqrt{D})$, $D > 1$ squarefree, so $\nu(K) = \nu(F)$ and every $\alpha \in \mathcal{O}_F^{\succ 0} \setminus \mathbb{Z}$ with $\tau(\alpha) < \nu(F)$ must have $2\alpha - c \in \mathcal{O}_F^2$ for some $c \in \{0, 1, \dots\}$ with $c < 2\min\sigma(\alpha)$. Take α to be the lower rung of the minimal vector, so $c \in \{0,1\}$ (Remark 2.3).

If $D \equiv 1 \pmod 4$, then $\nu = D/4$, the minimal vector is $\omega = \frac{1+\sqrt{D}}{2}$ up to $\pm\omega + k$, and $\alpha = \omega + u$ with $u = \lceil \frac{\sqrt{D}-1}{2} \rceil$ has $\tau(\alpha) = u + \frac{1}{2} < \frac{\sqrt{D}}{2} + 1 < \frac{D}{4}$ as soon as $D \ge 13$ (note $\min\sigma(\alpha) = u + \frac{1-\sqrt{D}}{2} \in (0,1]$). But $2\alpha - c = (2u + 1 - c) + \sqrt{D}$ is not a square in $\mathcal{O}_F$: if $\big(\frac{p+q\sqrt{D}}{2}\big)^2$ has $\sqrt{D}$-coefficient 1 then $pq = 2$, and then $p \not\equiv q \pmod 2$, so $\frac{p+q\sqrt{D}}{2} \notin \mathcal{O}_F$. Hence $D = 5$.

If $D \equiv 2, 3 \pmod 4$, then $\nu = D$, the minimal vector is $\sqrt{D}$, and $\alpha = \sqrt{D} + u$ with $u = \lfloor \sqrt{D} \rfloor + 1$ has $\tau(\alpha) = u < D$ for $D \ge 3$. A square $(p + q\sqrt{D})^2$ with $\sqrt{D}$-coefficient 2 has $pq = 1$, so $2\alpha - c = 2u - c + 2\sqrt{D}$ is a square only if $2u - c = D + 1$, i.e. $D = 2\lfloor\sqrt{D}\rfloor + 1 - c \le 2\sqrt{D} + 1$, i.e. $D \le 5$. This leaves $D = 2$ (where $\tau(\alpha) = \nu$) and $D = 3$ (where $2\alpha = (1+\sqrt{3})^2$ with $c = 0$). □

The three fields of Proposition 9.1 are exactly the quadratic members of $\mathcal{G}$. A direct computation (file `src/quadB.gp`) confirms that for squarefree $D \le 2000$ only $D = 2, 3, 5$ satisfy the half-square condition for *all* $\alpha \in \mathcal{O}_F^{\succ 0} \setminus \mathbb{Z}$ with $\tau(\alpha) < \nu(F)$. If $[K : F] = 2$ (a cubic F in a sextic K), the half-square condition no longer descends, but it is still strong: if some $\alpha \in \mathcal{O}_F^{\succ 0}$ with $\tau(\alpha) < \nu(F)$ has no half-square in F, then $K = F\big(\sqrt{2\alpha - c}\big)$ for one of the finitely many admissible c, so K is determined by F up to finitely many choices.

Proposition 9.2. *Let $K \in \mathcal{G}$ be sextic, with a minimal vector of Λ_K in a cubic subfield F. Then $\nu(F) = \nu(K) \le 9.58$ (both rungs cost at least $\frac{1}{2}\nu$, and the spread s of the minimal vector satisfies $s^2 \le 6\nu$), hence $d_F \le (3\nu)^3 \le 23{,}700$, and $t(F) \ge \frac{1}{2}\nu(F)$. Among the 214 cubic fields with $\nu(F) \le 9.58$ (all have $d_F \le 11777$), 143 violate $t(F) \ge \frac{1}{2}\nu(F)$; for 68 of them some $\alpha \in \mathcal{O}_F^{\succ 0}$ with $\tau(\alpha) < \nu(F)$ has no half-square in F, which forces $K = F(\sqrt{2\alpha - c})$ for an admissible c; the 92 fields so obtained all have certified witnesses. The remaining cubic fields, for which K is not determined, are exactly $d_F = 49$, $d_F = 81$, $d_F = 148$ (that is, $\mathbb{Q}(\zeta_7)^+$, $\mathbb{Q}(\zeta_9)^+$ and $\mathbb{Q}(\rho)$).*

Proposition 9.3. *No sextic $K \in \mathcal{G}$ is in regime 1 ($\nu_2 \ge 2\nu$) with a minimal vector generating K.*

Proof. By the discussion above K is generated by a root of one of the 9,785 polynomials that [14] excludes only by rigidity. They define 6429 distinct fields (2856 with 2 ramified), and each has a certified witness (file `results/regime1.txt`). □

Proposition 9.4. *No sextic $K \in \mathcal{G}$ has a minimal vector x_1 generating K with both rungs regular.*

Proof. If both rungs are regular, the downspread inequalities hold and $\nu \le 4.79$, so the enumeration of [14] without its quadratic filter (which needs $\tau \ge \nu$ for *all* totally positive α, not available in regime 2) contains the minimal polynomial of x_1. This enumeration (`src/enum6g.c`, `src/run_regime2X.sh`) has 178,438 leaves, exactly as in [14], and outputs 42,232 polynomials. Of the 16666 irreducible ones, 126 are excluded by BC0, 66 are excluded by HS, 1294 are excluded by NS, 5826 are excluded by R1, 9273 are excluded by UNR, 81 are excluded by W, where UNR means 2 unramified (Theorem 5.2), R1 means $\nu_2 \ge 2\nu$ (Proposition 9.3), NS that x_1 is not minimal in $\mathcal{O}_K$, BC0 an element with $\tau < \min(\nu, \frac{1}{2}\nu_2)$ (a witness by Corollary 2.2(4)), HS an element with $\tau < \nu$ and no half-square (Corollary 2.2(2)), and W a certified witness. □

Proposition 9.5. *No sextic $K \in \mathcal{G}$ has a minimal vector x_1 generating K with a half-square rung.*

Proof. After $x_1 \mapsto \pm x_1 + k$ we may assume that the lower rung is a half-square and $\tau(x_1) \in [-\frac{1}{2}, \frac{1}{2}]$. Then $y^2 = 2x_1 - C$ ($C = 2n + c$, $c \in \{0, 1\}$); if $\operatorname{Tr} x_1 = -a_1$ and $\sum \theta_j^2 = a_1^2 - 2a_2$, the power sums $p_2(y) = -2a_1 - 6C$ and $p_4(y) = 4(a_1^2 - 2a_2) + 4Ca_1 + 6C^2$ are fixed, the least y_j^2 lies in $(0, 2 - c]$, $\frac{1}{2}(\nu + c) \le \tau(x_1 - n) < \nu$, $\operatorname{Var}(y) \ge \nu$, both rungs of y and the upper rung of x_1 cost at least $\frac{1}{2}\nu$, and $\nu \le 15.75$; if $\nu > 7.77$ the upper rung is a half-square too, so $|y_j| < 2\sqrt{\nu}$. The enumeration of all such y (mode `GYL` of `src/enum6g.c`, validated against brute force in degree 4 for $\nu \le 8$ and in degree 6 for $\nu \le \frac{8}{3}$, see `src/validate/`) visits 422,110,544 leaves and returns 40,374 polynomials x_1. Of the 27907 irreducible ones, 13050 are excluded by BC0, 8174 are excluded by HS, 6667 are excluded by NS, 16 are excluded by W (notation as in Proposition 9.4). □

9.1. **Summary for degree six.**

Theorem 9.6. *Let $K \in \mathcal{G}$ be a totally real sextic field, and let x_1 be a minimal vector of Λ_K. Then 2 is ramified in K, K is not among the 9,815 sextic fields of the LMFDB list, the minimal vector x_1 does* not *generate K, and one of the following holds.*

(1) (Case B, quadratic) $F = \mathbb{Q}(x_1) \in \{\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5})\}$ *and* $\nu(K) = \nu(F) \in \{2, 3, \frac{5}{4}\}$.
(2) (Case B, cubic) $F = \mathbb{Q}(x_1) \in \{\mathbb{Q}(\zeta_7)^+, \mathbb{Q}(\zeta_9)^+, \mathbb{Q}(\rho)\}$, $[K : F] = 2$, *and* $\nu(K) = \nu(F) \in \{\frac{14}{9}, 2, \frac{20}{9}\}$.

Proof. 2 is ramified by Theorem 5.2, and K is not in the list by Theorem 4.1. Suppose $\mathbb{Q}(x_1) = K$. If both rungs of x_1 are regular, Proposition 9.4 gives a contradiction; otherwise some rung is a half-square (Remark 2.3) and Proposition 9.5 gives a contradiction. (Proposition 9.3 is the special case $\nu_2 \ge 2\nu$, used inside both.) Hence $\mathbb{Q}(x_1)$ is a proper subfield. If it is quadratic, $[K : \mathbb{Q}(x_1)] = 3$ is odd and Proposition 9.1 gives (1); if it is cubic, Proposition 9.2 gives (2). □

The members of $\mathcal{G}$ appearing in (1) and (2) are $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5}), \mathbb{Q}(\zeta_7)^+, \mathbb{Q}(\rho)$; only $\mathbb{Q}(\zeta_9)^+$ is not a member. So a sextic member of $\mathcal{G}$ would have to be an extension of degree 2 or 3 of one of these six fields (five of them members of $\mathcal{G}$) which inherits its variance minimum, with 2 ramified: precisely the "inherited" situation that the census identifies as extreme (Table 6).

Problem 9.7. Close the remaining case of Theorem 9.6.

Inherited minima. Bound the relative variance of K over $F \in \{\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3}), \mathbb{Q}(\sqrt{5})\}$ ($[K : F] = 3$) and over $F \in \{\mathbb{Q}(\zeta_7)^+, \mathbb{Q}(\zeta_9)^+, \mathbb{Q}(\rho)\}$ ($[K : F] = 2$) when 2 ramifies in K. The relative budget–cost of Proposition 2.4 then has a relative half-square alternative ($2\alpha - \gamma \in \mathcal{O}_K^2$ with $\gamma \in \Sigma\mathcal{O}_F^2$), which costs a factor 2 in the relative downspread bound; the relative enumerations of [14] have to be redone with that bound.

Together with Theorem 5.2 this would complete the classification of sextic members of $\mathcal{G}$.

10. Remarks on the working notes

We checked several statements of [10, 11] against independent computations.

(1) The values ν, ν_2, t for the members, for $\mathbb{Q}(\sqrt{13})$, and for the cyclic cubic fields of conductor $13, 91, 117, 133$ in [10, shape section] agree with ours.
(2) The table of excluded and surviving $m_0 \le 99$ in [11, table of $m_0 \le 99$] agrees with ours.
(3) The effective Northcott subsection of [10] states that the cubic scan with $d_K \le 3000$ leaves $\{49, 81, 169, 229, 257, 316, 321, 404, 469, 473, 564\}$ "together with the 2-ramified $d_K = 148$". This is the list of cubic fields with $d_K \le 3000$ and $\nu \le 3.7321$ irrespective of ramification; besides 148, the fields 316, 404, 564 are also 2-ramified and are therefore not covered by the 2-unramified threshold used there.
(4) The limiting-variance theorem of [10] gives $\sqrt{e}$ as the lower bound for $\liminf \mathrm{Var}$; Theorem 6.1 improves it.

11. Reproducibility

The repository contains:

- `src/gtest.gp`: the toolkit (field record, rigorous signs, the decision procedure `sosdecide`, unit criterion, totally positive generators modulo U_K^2, the witness search `gwitness2` with certification, and the invariants ν, ν_2, t);
- `src/census.gp`, `src/runcensus.sh`: the census (one output line per field: discriminant, polynomial, h^+/h, dyadic type, ν, status, witness, norm, time, ν_2, h, regulator, normalised shape minimum, trace of the witness);
- `src/run2o.gp`, `data/2o/`: $\mathcal{G}$-witnesses for the fields left by [14];
- `src/enum6.c` (copied from [14]), `src/varspec.sh`, `src/spectrum.gp`: the variance spectrum; `src/lp/varlp.py`, `src/lp/varlp_cert.py`: the linear programs and their interval-arithmetic verification;
- `src/cyclo.gp`: real cyclotomic fields; `src/regime1.gp`: the ramified sextic Case A test of §9; `src/gen_tables.py`: the tables of this note.

All runs used 4 cores; the full census takes about two hours, dominated by degree 8 and by the 193,467 cubic fields.

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